

EXPOSICION - INFORME DE EJECUTIVO

PROYECTO "APLICACIÓN MÓVIL COLABORATIVA PARA MEJORAR LA EFECTIVIDAD EN LA ASISTENCIA DE MASCOTAS PERDIDAS CON PARTICIPACIÓN DE VOLUNTARIOS EN LA CIUDAD DE TACNA, 2026"

Curso: Construcción de software II

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ALVARADO**

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RESUMEN EJECUTIVO

El proyecto SOS Mascota se encuentra actualmente en la fase de construcción e implementación, con un avance general del 45%. Se han completado al 100% el levantamiento de requerimientos (SRS) y el diseño arquitectónico (SAD), estableciendo una base sólida para el desarrollo. En esta etapa ya se han implementado 5 de los 17 requerimientos funcionales y se han aprobado 93 pruebas automatizadas, correspondientes a las funcionalidades desarrolladas.



2. OBJETIVOS DEL PROYECTO

- Desarrollar e implementar una aplicación móvil colaborativa que permita optimizar el proceso de búsqueda, reporte y localización de mascotas perdidas en la ciudad de Tacna, con participación activa de voluntarios.

3. OBJETIVOS ESPECÍFICOS

- Autenticar y registrar usuarios (usuarios, voluntarios, y administrador).
- Permitir el registro de reportes de mascotas perdidas con foto, descripción, contacto y ubicación.
- Permitir el registro de avistamientos y su asociación automática a reportes existentes mediante reconocimiento de imágenes con IA.
- Visualizar mascotas reportadas en un listado y en un mapa interactivo.
- Facilitar la comunicación entre usuarios mediante notificaciones push y una sección de comentarios.
- Proveer paneles especializados de administración: gestión de usuarios, gestión de reportes y estadísticas avanzadas.
- Garantizar la calidad del código y del producto mediante revisión de código y un catálogo formal de pruebas.

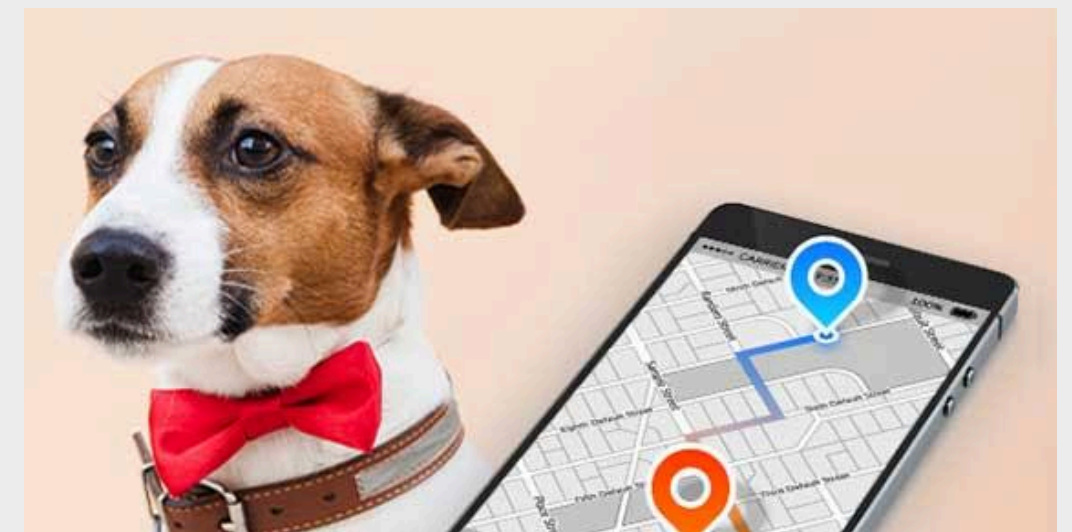
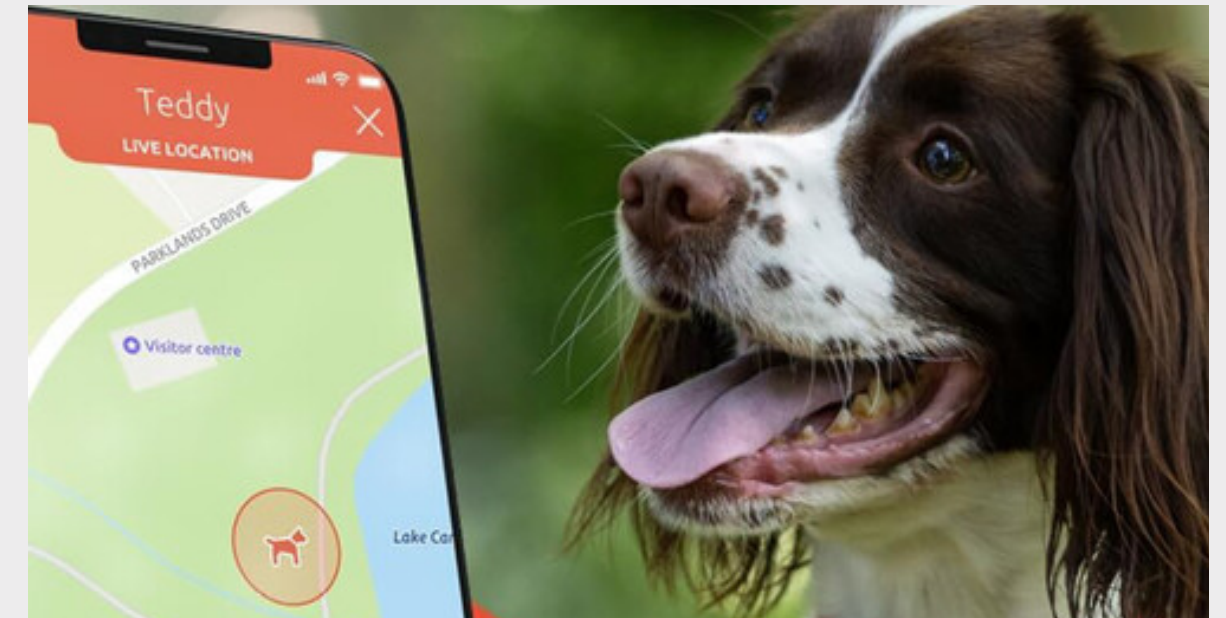


4. ALCANCE

- El proyecto comprende el análisis, diseño, desarrollo, pruebas, revisión de código e implementación de la aplicación móvil SOS Mascota para dispositivos Android, desarrollada en Flutter bajo el patrón MVVM y soportada en Firebase (Authentication, Firestore Storage, Cloud Messaging).

PRINCIPALES FUNCIONALIDADES (17 REQUERIMIENTOS FUNCIONALES, RF-01 A RF-17)

- Autenticación, registro y recuperación de contraseña (RF-01, RF-02, RF-03).
- Registro de reportes de mascotas perdidas y de avistamientos, con validación de imágenes por IA (RF-04, RF-05).
- Mapa interactivo con reportes y avistamientos georreferenciados (RF-06).
- Notificaciones automáticas y comunicación entre usuarios coincidentes (RF-07, RF-08).
- Gestión de perfil, listado público de mascotas y gestión de reportes propios (RF-09, RF-10, RF-11).
- Sección comunitaria de comentarios y cierre de sesión (RF-12, RF-13).
- Panel administrativo: gestión del panel, usuarios, reportes y estadísticas avanzadas (RF-14 a RF-17).



5.AVANCE DEL PROYECTO

Área	Avance	Estado
Levantamiento de requerimientos (SRS v3.0)	100%	Completado
Análisis y diseño arquitectónico (SAD, modelo C4)	100%	Completado
Desarrollo (código productivo con pruebas unitarias aprobadas: 5 de 17 RF)	≈ 29%	En progreso
Pruebas (catálogo de casos: 51 de 141 filas aprobadas)	≈ 36%	En progreso
Revisión de código (Lab 01-02: autenticación, avistamientos, comentarios)	≈ 42%	En progreso
Implementación / despliegue a producción	0%	Pendiente
Capacitación a usuarios	0%	Pendiente
Avance general estimado	≈ 45%	En progreso

5.1. EVIDENCIA DE PRUEBAS EN CONSOLA:

PRUEBA 01- TEST/RF01_LOGIN

```
Problems Output Debug Console Terminal Ports
PS C:\Users\Usuario\Downloads\ProyectoUnidad02_Construccion\S05Mascota_V1\SosMascota> flutter test test/rf01_login_test.dart -v
[ +4 ms] Artifact Instance of 'GradleWrapper' is not required, skipping update.
[ +1 ms] Artifact Instance of 'AndroidGenSnapshotArtifacts' is not required, skipping update.
[       ] Artifact Instance of 'AndroidInternalBuildArtifacts' is not required, skipping update.
[       ] Artifact Instance of 'IOSEngineArtifacts' is not required, skipping update.
[       ] Artifact Instance of 'FlutterWebSdk' is not required, skipping update.
[ +2 ms] Artifact Instance of 'LegacyCanvasKitRemover' is not required, skipping update.
[       ] Artifact Instance of 'FlutterSdk' is not required, skipping update.
[       ] Artifact Instance of 'WindowsEngineArtifacts' is not required, skipping update.
[       ] Artifact Instance of 'MacOSEngineArtifacts' is not required, skipping update.
[       ] Artifact Instance of 'LinuxEngineArtifacts' is not required, skipping update.
[       ] Artifact Instance of 'LinuxFuchsiaSDKArtifacts' is not required, skipping update.
[       ] Artifact Instance of 'MacOSFuchsiaSDKArtifacts' is not required, skipping update.
[ +1 ms] Artifact Instance of 'FlutterRunnerSDKArtifacts' is not required, skipping update.
[       ] Artifact Instance of 'FlutterRunnerDebugSymbols' is not required, skipping update.
[       ] Artifact Instance of 'IosUsbArtifacts' is not required, skipping update.
[       ] Artifact Instance of 'IosUsbArtifacts' is not required, skipping update.
[       ] Artifact Instance of 'IosUsbArtifacts' is not required, skipping update.
[       ] Artifact Instance of 'IosUsbArtifacts' is not required, skipping update.
[ +1 ms] Artifact Instance of 'IosUsbArtifacts' is not required, skipping update.
[ +1 ms] Artifact Instance of 'IosUsbArtifacts' is not required, skipping update.
[       ] Artifact Instance of 'FontSubsetArtifacts' is not required, skipping update.
[       ] Artifact Instance of 'PubDependencies' is not required, skipping update.
[+175 ms] Found 1 files which will be executed as Widget Tests.
[ +7 ms] Artifact Instance of 'AndroidGenSnapshotArtifacts' is not required, skipping update.
[ +1 ms] Artifact Instance of 'AndroidInternalBuildArtifacts' is not required, skipping update.
[ +1 ms] Artifact Instance of 'IOSEngineArtifacts' is not required, skipping update.
[ +1 ms] Artifact Instance of 'FlutterWebSdk' is not required, skipping update.
[       ] Artifact Instance of 'FlutterEngineStamp' is not required, skipping update.
[       ] Artifact Instance of 'LegacyCanvasKitRemover' is not required, skipping update.
[ +4 ms] Artifact Instance of 'WindowsEngineArtifacts' is not required, skipping update.
[ +1 ms] Artifact Instance of 'MacOSEngineArtifacts' is not required, skipping update.
[       ] Artifact Instance of 'LinuxEngineArtifacts' is not required, skipping update.
[       ] Artifact Instance of 'LinuxFuchsiaSDKArtifacts' is not required, skipping update.
[ +1 ms] Artifact Instance of 'MacOSFuchsiaSDKArtifacts' is not required, skipping update.
[ +1 ms] Artifact Instance of 'FlutterRunnerSDKArtifacts' is not required, skipping update.
```

```
Problems Output Debug Console Terminal Ports
PS C:\Users\Usuario\Downloads\ProyectoUnidad02_Construccion\S05Mascota_V1\SosMascota> flutter test test/rf01_login_test.dart -v
[ +12 ms] test 0: Started flutter_tester process at pid 32068
[ +357 ms] test 0: connected to test device, now awaiting test result
[ +1 ms] test 0: Waiting for test harness or tests to finish
00:03 +12: RF-01 ... Usuario - Validación de formulario ✖ ERROR: código desconocido retorna el message del SDK
[ +181 ms] test 0: Test harness is no longer needed by test process
[       ] test 0: finished
[ +1 ms] test 0: cleaning up...
[       ] test 0: ensuring test device is terminated.
[       ] test 0: Terminating flutter_tester process
[ +1 ms] test 0: Shutting down DevTools server
[       ] test 0: Test process is no longer needed by test harness
[ +2 ms] test 0: Shutting down test harness socket server
[ +12 ms] test 0: flutter_tester process at pid 32068 exited with code=-1
[       ] test 0: deleting temporary directory
[ +5 ms] test 0: finished
00:03 +12: All tests passed!
[ +9 ms] Deleting C:\Users\Usuario\AppData\Local\Temp\flutter_tools.a1450f3e\Flutter_test_compiler.5733aa60...
[ +4 ms] killing pid 36512
[ +41 ms] Deleting C:\Users\Usuario\AppData\Local\Temp\flutter_tools.a1450f3e\Flutter_test_fonts.ea38d99d...
[ +3 ms] test package returned with exit code 0
[ +1 ms] Runtime for phase TestRunner: Wall-clock: 0:00:03.652020; combined: 0:00:03.652082.
[       ] Runtime for phase Compile: Wall-clock: 0:00:02.900491; combined: 0:00:02.500514.
[       ] Runtime for phase Run: Wall-clock: 0:00:00.570049; combined: 0:00:00.570051.
[       ] Runtime for phase CoverageTotal: Wall-clock: 0:00:00.000000; combined: 0:00:00.000000.
[       ] Runtime for phase CoverageCollect: Wall-clock: 0:00:00.000000; combined: 0:00:00.000000.
[       ] Runtime for phase CoverageParseJson: Wall-clock: 0:00:00.000000; combined: 0:00:00.000000.
[       ] Runtime for phase CoverageAddHitmap: Wall-clock: 0:00:00.000000; combined: 0:00:00.000000.
[       ] Runtime for phase CoverageDataCollect: Wall-clock: 0:00:00.000000; combined: 0:00:00.000000.
[       ] Runtime for phase WatcherFinishedTest: Wall-clock: 0:00:00.000730; combined: 0:00:00.000739.
[ +1 ms] "flutter test" took 5.393ms.
[+107 ms] Running 3 shutdown hooks
[ +3 ms] Shutdown hooks complete
[ +64 ms] exiting with code 0
PS C:\Users\Usuario\Downloads\ProyectoUnidad02_Construccion\S05Mascota_V1\SosMascota>
```

5.1. EVIDENCIA DE PRUEBAS EN CONSOLA:

PRUEBA 02- TEST/RF02_REGISTRO

```
Problems Output Debug Console Terminal Ports
[ +64 ms] exiting with code 0
PS C:\Users\Usuario\Downloads\ProyectoUnidad02_Construccion\S05Mascota_V1\SosMascota> flutter test test/rf02_registro_test.dart -v
[ +442 ms] Artifact Instance of 'MaterialFonts' is not required, skipping update.
[ +2 ms] Artifact Instance of 'GradleWrapper' is not required, skipping update.
[ ] Artifact Instance of 'AndroidGenSnapshotArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'AndroidInternalBuildArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'IOSEngineArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'FlutterWebSdk' is not required, skipping update.
[ +1 ms] Artifact Instance of 'LegacyCanvasKitRemover' is not required, skipping update.
[ ] Artifact Instance of 'FlutterSdk' is not required, skipping update.
[ ] Artifact Instance of 'WindowsEngineArtifacts' is not required, skipping update.
[ +1 ms] Artifact Instance of 'MacOSEngineArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'LinuxEngineArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'LinuxFuchsiaSDKArtifacts' is not required, skipping update.
[ +1 ms] Artifact Instance of 'MacOSFuchsiaSDKArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'FlutterRunnerSDKArtifacts' is not required, skipping update.
[ +1 ms] Artifact Instance of 'FlutterRunnerDebugSymbols' is not required, skipping update.
[ ] Artifact Instance of 'IosUsbArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'IosUsbArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'IosUsbArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'IosUsbArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'IosUsbArtifacts' is not required, skipping update.
[ +1 ms] Artifact Instance of 'FontSubsetArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'PubDependencies' is not required, skipping update.
[ +135 ms] Found 1 files which will be executed as Widget Tests.
[ +6 ms] Artifact Instance of 'AndroidGenSnapshotArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'AndroidInternalBuildArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'IOSEngineArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'FlutterWebSdk' is not required, skipping update.
[ ] Artifact Instance of 'FlutterEngineStamp' is not required, skipping update.
[ ] Artifact Instance of 'LegacyCanvasKitRemover' is not required, skipping update.
[ +4 ms] Artifact Instance of 'WindowsEngineArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'MacOSEngineArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'LinuxEngineArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'LinuxFuchsiaSDKArtifacts' is not required, skipping update.
```

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```
Problems Output Debug Console Terminal Ports
PS C:\Users\Usuario\Downloads\ProyectoUnidad02_Construccion\S05Mascota_V1\SosMascota> flutter test test/rf02_registro_test.dart -v
00:03 +0: ...Downloads/ProyectoUnidad02_Construccion/S05Mascota_V1/SosMascota/test/rf02_registro_test.dart [
+586 ms] test 0: connected to test device, now awaiting test result
[ +3 ms] test 0: Waiting for test harness or tests to finish
00:03 +20: RF-02 ... - Modelo Usuario serialización ✓ (ÉXITO: fromMap() usa valores por defecto si faltan campos
[
+631 ms] test 0: Test harness is no longer needed by test process
[ +2 ms] test 0: finished
[ +4 ms] test 0: cleaning up...
[ +1 ms] test 0: ensuring test device is terminated.
[ +2 ms] test 0: Terminating flutter_tester process
[ +5 ms] test 0: Shutting down DevTools server
[ +2 ms] test 0: Test process is no longer needed by test harness
[ +9 ms] test 0: Shutting down test harness socket server
[ +41 ms] test 0: flutter_tester process at pid 22876 exited with code--1
[ +2 ms] test 0: deleting temporary directory
[ +10 ms] test 0: finished
00:03 +20: All tests passed!
[ +14 ms] Deleting C:\Users\Usuario\AppData\Local\Temp\flutter_tools.ac81a845\flutter_test_compiler.fbc6dea0...
[ +10 ms] killing pid 12684
[ +89 ms] Deleting C:\Users\Usuario\AppData\Local\Temp\flutter_tools.ac81a845\flutter_test_fonts.6647568f...
[ +7 ms] test package returned with exit code 0
[ +2 ms] Runtime for phase TestRunner: Wall-clock: 0:00:04.231111; combined: 0:00:04.231149.
[ +1 ms] Runtime for phase Compile: Wall-clock: 0:00:02.344900; combined: 0:00:02.344923.
[ +1 ms] Runtime for phase Run: Wall-clock: 0:00:01.259379; combined: 0:00:01.259382.
[ ] Runtime for phase CoverageTotal: Wall-clock: 0:00:00.000000; combined: 0:00:00.000000.
[ ] Runtime for phase CoverageCollect: Wall-clock: 0:00:00.000000; combined: 0:00:00.000000.
[ +1 ms] Runtime for phase CoverageParseJson: Wall-clock: 0:00:00.000000; combined: 0:00:00.000000.
[ +1 ms] Runtime for phase CoverageAddHitmap: Wall-clock: 0:00:00.000000; combined: 0:00:00.000000.
[ +1 ms] Runtime for phase CoverageDataCollect: Wall-clock: 0:00:00.000000; combined: 0:00:00.000000.
[ ] Runtime for phase WatcherFinishedTest: Wall-clock: 0:00:00.002326; combined: 0:00:00.002330.
[ +1 ms] "flutter test" took 6.483ms.
[ +255 ms] Running 3 shutdown hooks
[ +9 ms] Shutdown hooks complete
[ +74 ms] exiting with code 0
PS C:\Users\Usuario\Downloads\ProyectoUnidad02_Construccion\S05Mascota_V1\SosMascota>
```

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5.1. EVIDENCIA DE PRUEBAS EN CONSOLA:

PRUEBA 03- TEST/RF04_REPORTE

```
Problems Output Debug Console Terminal Ports
[ +74 ms] exiting with code 0
PS C:\Users\Usuario\Downloads\ProyectoUnidad02_Construccion\SOSMascota_V1\SosMascota> flutter test test/rf04_reporte_test.dart -v
[ +276 ms] Artifact Instance of 'MaterialFonts' is not required, skipping update.
[ +2 ms] Artifact Instance of 'GradleWrapper' is not required, skipping update.
[ ] Artifact Instance of 'AndroidGenSnapshotArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'AndroidInternalBuildArtifacts' is not required, skipping update.
[ +1 ms] Artifact Instance of 'IOSEngineArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'FlutterWebSdk' is not required, skipping update.
[ +1 ms] Artifact Instance of 'LegacyCanvasKitRemover' is not required, skipping update.
[ ] Artifact Instance of 'FlutterSdk' is not required, skipping update.
[ +1 ms] Artifact Instance of 'WindowsEngineArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'MacOSEngineArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'LinuxEngineArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'LinuxFuchsiaSDKArtifacts' is not required, skipping update.
[ +2 ms] Artifact Instance of 'MacOSFuchsiaSDKArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'FlutterRunnerSDKArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'FlutterRunnerDebugSymbols' is not required, skipping update.
[ ] Artifact Instance of 'IosUsbArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'IosUsbArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'IosUsbArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'IosUsbArtifacts' is not required, skipping update.
[ +1 ms] Artifact Instance of 'IosUsbArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'IosUsbArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'FontSubsetArtifacts' is not required, skipping update.
[ +1 ms] Artifact Instance of 'PubDependencies' is not required, skipping update.
[ +141 ms] Found 1 files which will be executed as Widget Tests.
[ +6 ms] Artifact Instance of 'AndroidGenSnapshotArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'AndroidInternalBuildArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'IOSEngineArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'FlutterWebSdk' is not required, skipping update.
[ ] Artifact Instance of 'FlutterEngineStamp' is not required, skipping update.
[ ] Artifact Instance of 'LegacyCanvasKitRemover' is not required, skipping update.
[ +5 ms] Artifact Instance of 'WindowsEngineArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'MacOSEngineArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'LinuxEngineArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'LinuxFuchsiaSDKArtifacts' is not required, skipping update.
```

```
Problems Output Debug Console Terminal Ports
PS C:\Users\Usuario\Downloads\ProyectoUnidad02_Construccion\SOSMascota_V1\SosMascota> flutter test test/rf04_reporte_test.dart -v
[ +10 ms] test 0: Started flutter_tester process at pid 36932
[ +305 ms] test 0: connected to test device, now awaiting test result
[ +1 ms] test 0: Waiting for test harness or tests to finish
00:04 +20: RF-04 ... de campos requeridos ✗ ERROR: reporte sin latitud/longitud tiene coordenadas nulas
[ +244 ms] test 0: Test harness is no longer needed by test process
[ +1 ms] test 0: finished
[ +2 ms] test 0: cleaning up...
[ +1 ms] test 0: ensuring test device is terminated.
[ ] test 0: Terminating flutter_tester process
[ +1 ms] test 0: Shutting down DevTools server
[ ] test 0: Test process is no longer needed by test harness
[ +3 ms] test 0: Shutting down test harness socket server
[ +15 ms] test 0: flutter_tester process at pid 36932 exited with code=-1
[ +2 ms] test 0: deleting temporary directory
[ +7 ms] test 0: finished
00:04 +20: All tests passed!
[ +23 ms] Deleting C:\Users\Usuario\AppData\Local\Temp\flutter_tools.3864ba1a\flutter_test_compiler.903223c8...
[ +5 ms] killing pid 26104
[ +55 ms] Deleting C:\Users\Usuario\AppData\Local\Temp\flutter_tools.3864ba1a\flutter_test_fonts.e7f69162...
[ +3 ms] test package returned with exit code 0
[ +1 ms] Runtime for phase TestRunner: Wall-clock: 0:00:05.022790; combined: 0:00:05.022824.
[ ] Runtime for phase Compile: Wall-clock: 0:00:03.937154; combined: 0:00:03.937181.
[ ] Runtime for phase Run: Wall-clock: 0:00:00.635765; combined: 0:00:00.635768.
[ ] Runtime for phase CoverageTotal: Wall-clock: 0:00:00.000000; combined: 0:00:00.000000.
[ ] Runtime for phase CoverageCollect: Wall-clock: 0:00:00.000000; combined: 0:00:00.000000.
[ ] Runtime for phase CoverageParseJson: Wall-clock: 0:00:00.000000; combined: 0:00:00.000000.
[ ] Runtime for phase CoverageAddHitmap: Wall-clock: 0:00:00.000000; combined: 0:00:00.000000.
[ ] Runtime for phase CoverageDataCollect: Wall-clock: 0:00:00.000000; combined: 0:00:00.000000.
[ ] Runtime for phase WatcherFinishedTest: Wall-clock: 0:00:00.001297; combined: 0:00:00.001299.
[ ] "flutter test" took 7.134ms.
[ +127 ms] Running 3 shutdown hooks
[ +5 ms] Shutdown hooks complete
[ +69 ms] exiting with code 0
PS C:\Users\Usuario\Downloads\ProyectoUnidad02_Construccion\SOSMascota_V1\SosMascota>
```

5.1. EVIDENCIA DE PRUEBAS EN CONSOLA:

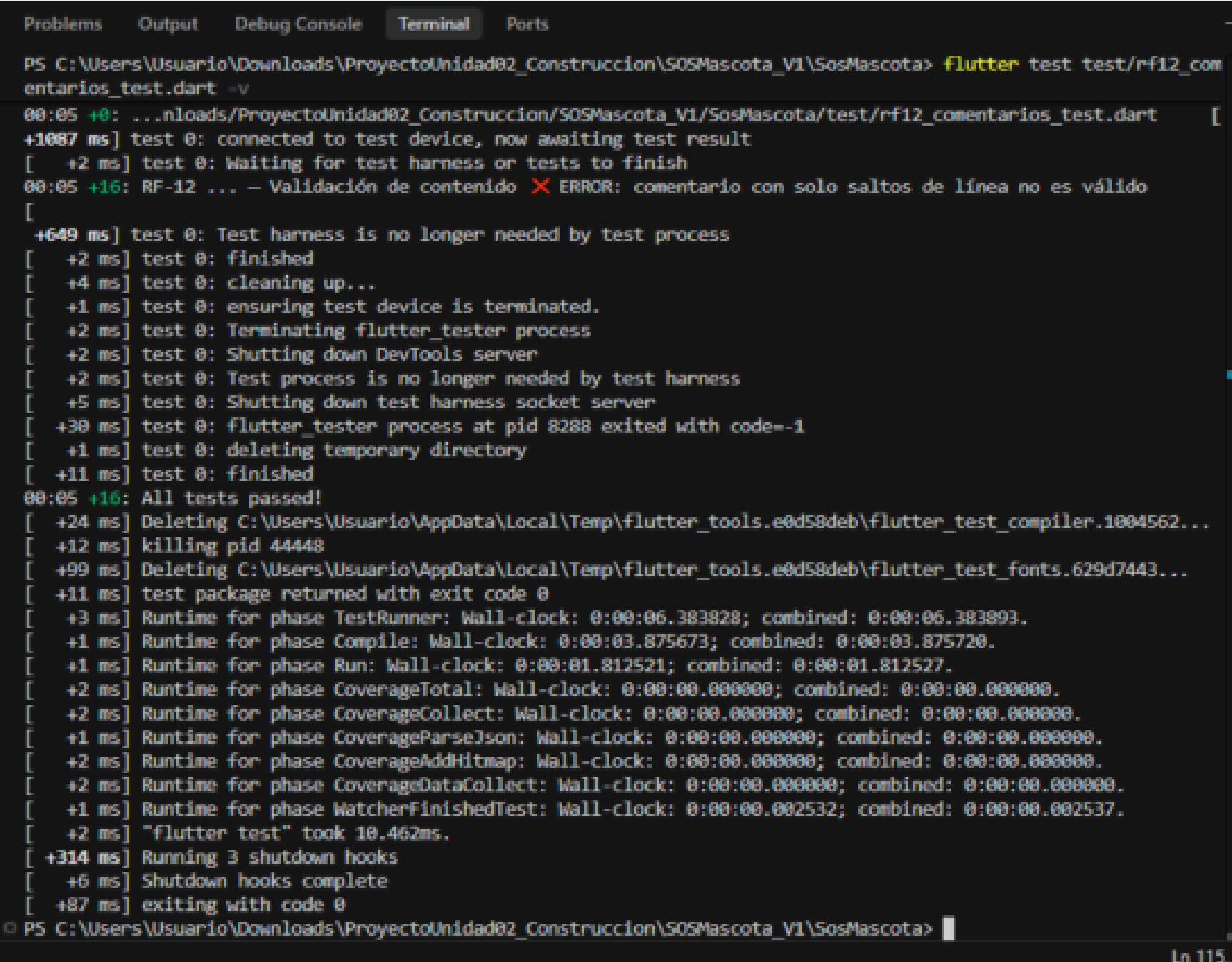
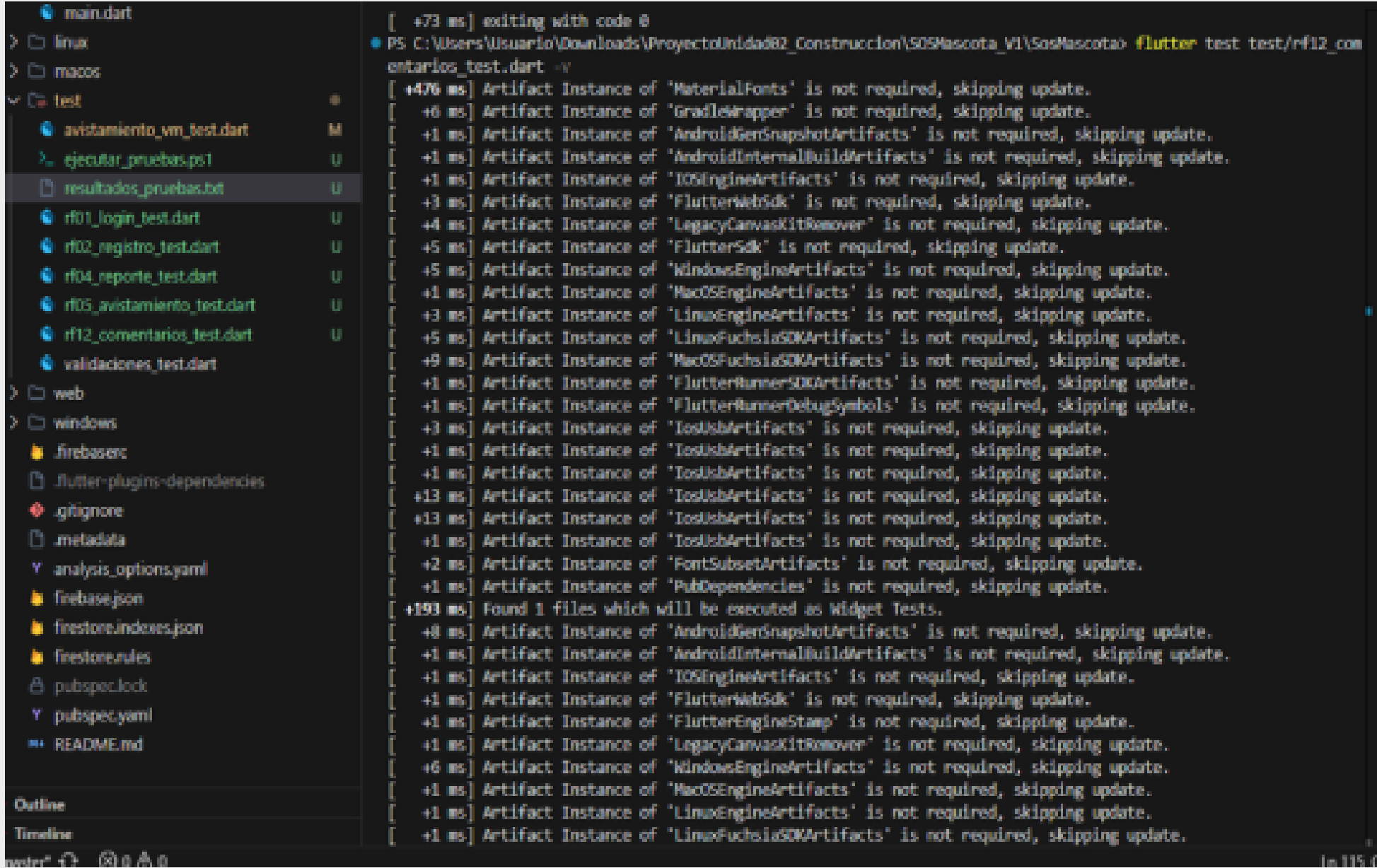
PRUEBA 04- TEST/RF05_AVISTAMIENTO

```
Problems Output Debug Console Terminal Ports
[ +5 ms] Shutdown hooks complete
[ +69 ms] exiting with code 0
PS C:\Users\Usuario\Downloads\ProyectoUnidad02_Construccion\S05Mascota_V1\SosMascota> flutter test test/rf05_avistamiento_test.dart -v
[ +300 ms] Artifact Instance of 'MaterialFonts' is not required, skipping update.
[ +2 ms] Artifact Instance of 'GradleWrapper' is not required, skipping update.
[ ] Artifact Instance of 'AndroidGenSnapshotArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'AndroidInternalBuildArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'IOSEngineArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'FlutterWebSdk' is not required, skipping update.
[ +1 ms] Artifact Instance of 'LegacyCanvasKitRemover' is not required, skipping update.
[ ] Artifact Instance of 'FlutterSdk' is not required, skipping update.
[ +1 ms] Artifact Instance of 'WindowsEngineArtifacts' is not required, skipping update.
[ +1 ms] Artifact Instance of 'MacOSEngineArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'LinuxEngineArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'LinuxFuchsiaSDKArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'MacOSFuchsiaSDKArtifacts' is not required, skipping update.
[ +1 ms] Artifact Instance of 'FlutterRunnerSDKArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'FlutterRunnerDebugSymbols' is not required, skipping update.
[ ] Artifact Instance of 'IosUsbArtifacts' is not required, skipping update.
[ +1 ms] Artifact Instance of 'IosUsbArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'IosUsbArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'IosUsbArtifacts' is not required, skipping update.
[ +1 ms] Artifact Instance of 'IosUsbArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'IosUsbArtifacts' is not required, skipping update.
[ +1 ms] Artifact Instance of 'FontSubsetArtifacts' is not required, skipping update.
[ +1 ms] Artifact Instance of 'PubDependencies' is not required, skipping update.
[ +155 ms] Found 1 files which will be executed as Widget Tests.
[ +7 ms] Artifact Instance of 'AndroidGenSnapshotArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'AndroidInternalBuildArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'IOSEngineArtifacts' is not required, skipping update.
[ +1 ms] Artifact Instance of 'FlutterWebSdk' is not required, skipping update.
[ +1 ms] Artifact Instance of 'FlutterEngineStamp' is not required, skipping update.
[ +1 ms] Artifact Instance of 'LegacyCanvasKitRemover' is not required, skipping update.
[ +5 ms] Artifact Instance of 'WindowsEngineArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'MacOSEngineArtifacts' is not required, skipping update.
[ ] Artifact Instance of 'LinuxEngineArtifacts' is not required, skipping update.
```

```
Problems Output Debug Console Terminal Ports
PS C:\Users\Usuario\Downloads\ProyectoUnidad02_Construccion\S05Mascota_V1\SosMascota> flutter test test/rf05_avistamiento_test.dart -v
00:03 +0: ...loads/ProyectoUnidad02_Construccion/S05Mascota_V1/SosMascota/test/rf05_avistamiento_test.dart [
+572 ms] test 0: connected to test device, now awaiting test result
[ +2 ms] test 0: Waiting for test harness or tests to finish
00:03 +17: RF-05 ... Modelo Avistamiento serialización ✖ ERROR: avistamiento sin descripción tiene campo vacío
[
+286 ms] test 0: Test harness is no longer needed by test process
[ +1 ms] test 0: finished
[ +2 ms] test 0: cleaning up...
[ +1 ms] test 0: ensuring test device is terminated.
[ +1 ms] test 0: Terminating flutter_tester process
[ +1 ms] test 0: Shutting down DevTools server
[ +1 ms] test 0: Test process is no longer needed by test harness
[ +3 ms] test 0: Shutting down test harness socket server
[ +19 ms] test 0: flutter_tester process at pid 44860 exited with code=-1
[ +1 ms] test 0: deleting temporary directory
[ +7 ms] test 0: finished
00:03 +17: All tests passed!
[ +11 ms] Deleting C:\Users\Usuario\AppData\Local\Temp\flutter_tools.d8edc147\flutter_test_compiler.89ff78cb...
[ +7 ms] killing pid 12808
[ +64 ms] Deleting C:\Users\Usuario\AppData\Local\Temp\flutter_tools.d8edc147\flutter_test_fonts.2997194e...
[ +4 ms] test package returned with exit code 0
[ +1 ms] Runtime for phase TestRunner: Wall-clock: 0:00:03.858799; combined: 0:00:03.858837.
[ ] Runtime for phase Compile: Wall-clock: 0:00:02.475237; combined: 0:00:02.475256.
[ ] Runtime for phase Run: Wall-clock: 0:00:00.896689; combined: 0:00:00.896612.
[ ] Runtime for phase CoverageTotal: Wall-clock: 0:00:00.000000; combined: 0:00:00.000000.
[ ] Runtime for phase CoverageCollect: Wall-clock: 0:00:00.000000; combined: 0:00:00.000000.
[ ] Runtime for phase CoverageParseJson: Wall-clock: 0:00:00.000000; combined: 0:00:00.000000.
[ ] Runtime for phase CoverageAddHitmap: Wall-clock: 0:00:00.000000; combined: 0:00:00.000000.
[ ] Runtime for phase CoverageDataCollect: Wall-clock: 0:00:00.000000; combined: 0:00:00.000000.
[ ] Runtime for phase WatcherFinishedTest: Wall-clock: 0:00:00.001023; combined: 0:00:00.001024.
[ +1 ms] "flutter test" took 6.523ms.
[ +149 ms] Running 3 shutdown hooks
[ +5 ms] Shutdown hooks complete
[ +70 ms] exiting with code 0
PS C:\Users\Usuario\Downloads\ProyectoUnidad02_Construccion\S05Mascota_V1\SosMascota>
```

5.1. EVIDENCIA DE PRUEBAS EN CONSOLA:

PRUEBA 05- TEST/RF12_COMENTARIOS



6. PRINCIPALES ENTREGABLES

Entregable	Estado
Documento de Especificación de Requerimientos de Software (SRS v3.0)	Completado
Informe de Arquitectura de Software (SAD)	Completado
Guía de revisión de código (Lab 01 y Lab 02 - Especificaciones Funcionales)	En curso
Catálogo de casos de prueba (v5)	En curso
Matriz de trazabilidad SRS – SAD – Implementación	En curso (4 de 17 RF documentados)
Código fuente de la aplicación (Flutter / Firebase)	Parcial (5 de 17 RF)
Manual de usuario	Pendiente
Aplicación desplegada en ambiente Beta/producción	Pendiente

7. RIESGOS Y PROBLEMAS IDENTIFICADOS

Riesgo / Problema	Impacto	Acción propuesta
Divergencia entre RF especificados (17) y RF con código y pruebas aprobadas (5)	Alto	Priorizar el desarrollo de los RF de alta prioridad restantes (RF-06 Mapa, RF-07 Notificaciones, RF-10 Listado público) antes del cierre del curso.
Hallazgo en RF-08 (chat): se detectó una implementación real de chat con Firestore (pantalla_chat.dart, pantalla_chats_activos.dart) que contradice la decisión de alcance de excluir el chat en v1.0. El equipo optó por sustituirla por un stub (pantalla_chat_stub.dart) que redirige a "Disponible en la próxima versión"	Medio	Verificar manualmente el nuevo stub, documentar formalmente la decisión (Opción B) en el SRS/SAD y actualizar el catálogo de pruebas del Caso 8, cuyo resultado "Aprobado" previo ya no es válido.
Alto volumen de pruebas pendientes: 87 de 141 filas del catálogo (integración, no funcionales, regresión, humo y punto a punto) aún no se han ejecutado	Medio	Programar la ejecución de las pruebas pendientes conforme cada RF avance a estado "Desarrollo completado".
Inconsistencia entre el diseño (SAD indica Google Maps API como proveedor de mapas) y la implementación real (el código usa OpenStreetMap + Nominatim vía flutter_map)	Medio	Actualizar el SAD para reflejar el proveedor de mapas realmente utilizado, o migrar la implementación a Google Maps API si así lo exige el alcance aprobado.
Plazo ajustado: quedan aproximadamente 11 semanas (al 04/12/2026) para completar 12 RF restantes, elevar la cobertura de pruebas, capacitar usuarios e implementar	Alto	Definir un cronograma semanal de cierre con hitos de control, priorizando RF de prioridad Alta sobre los de prioridad Baja.

7.1. EVIDENCIA:

PANTALLA_CHAT_ACTIVOS.DART

```
Explorer
  ▼ SosMascota
    > .dart_tool
    > .firebase
    > android
    > assets
    > build
    > ios
    ▼ lib
      > modelo
      > servicios
      > utils
      ▼ vista
        > admin
        > auth
        ▼ chat
          pantalla_chat.dart
          pantalla_chats_activos.dart
        > maps
        > reportes
        > usuario
        > vistamodelo
        > widgets
      app.dart
      firebase_options.dart
      main.dart
    > linux
    > macos
    > test
    > web
    > windows
    .firebase
    .flutter-plugins-dependencies
    .gitignore
    .metadata
    analysis_options.yaml
    firebase.json
    firestore.indexes.json
    firestore.rules
```

```
lib > vista > chat > pantalla_chats_activos.dart
1  import 'dart:math';
2  import 'package:cloud_firestore/cloud_firestore.dart';
3  import 'package:firebase_auth/firebase_auth.dart';
4  import 'package:flutter/material.dart';
5  import 'pantalla_chat.dart';
6
7  class PantallaChatsActivos extends StatefulWidget {
8    const PantallaChatsActivos({super.key});
9
10   @override
11   State<PantallaChatsActivos> createState() => _PantallaChatsActivosState();
12 }
13
14 class _PantallaChatsActivosState extends State<PantallaChatsActivos> {
15   final List<String> avataresDefault = [
16     'assets/avatars/cat.png',
17     'assets/avatars/dog.png',
18     'assets/avatars/bear.png',
19   ];
20
21   final Map<String, String> avatarAsignado = {};
22
23   String obtenerAvatar(String userId) {
24     if (!avatarAsignado.containsKey(userId)) {
25       final random = Random();
26       avatarAsignado[userId] =
27         avataresDefault[random.nextInt(avataresDefault.length)];
28     }
29     return avatarAsignado[userId]!;
30   }
31
32   String _formatearHora(Timestamp? ts) {
33     if (ts == null) return '';
34     final dt = ts.toDate();
35     final ahora = DateTime.now();
36     final diff = ahora.difference(dt);
37
38     if (diff.inDays == 0) {
39       return '${dt.hour.toString().padLeft(2, '0')}:${dt.minute.toString().padLeft(2, '0')}';
40     }
41     if (diff.inDays == 1) return 'Ayer';
42     if (diff.inDays < 7) return '${diff.inDays}d';
43     return '${dt.day}/${dt.month}';
```

```
93
94   if (snapshot.connectionState == ConnectionState.waiting && !snapshot.hasData) {
95     return const Center(child: CircularProgressIndicator());
96   }
97
98   // Ordenar client-side por fechaInicio descendente
99   final chats = (snapshot.data?.docs ?? []).sort((a, b) {
100     final dataA = a.data() as Map<String, dynamic>;
101     final dataB = b.data() as Map<String, dynamic>;
102     final tsA = dataA['fechaInicio'] as Timestamp?;
103     final tsB = dataB['fechaInicio'] as Timestamp?;
104     if (tsA == null && tsB == null) return 0;
105     if (tsA == null) return 1;
106     if (tsB == null) return -1;
107     return tsB.compareTo(tsA);
108   });
109
110   if (chats.isEmpty) {
111     return _buildEstadoVacio();
112   }
113
114   return ListView.builder(
115     padding: const EdgeInsets.symmetric(vertical: 8, horizontal: 12),
116     itemCount: chats.length,
117     itemBuilder: (context, i) {
118       final chat = chats[i];
119       final data = chat.data() as Map<String, dynamic>;
120
121       final isPublicador = data['publicadorId'] == currentUser.uid;
122       final otherUserId =
123         !isPublicador ? data['usuarioId'] : data['publicadorId'];
124       final tipo = data['tipo'] ?? 'reporte';
125
126       return FutureBuilder<DocumentSnapshot>({
127         future: FirebaseFirestore.instance
128           .collection('usuarios')
129           .doc(otherUserId)
130           .get(),
131         builder: (context, userSnap) {
132           if (userSnap.hasData) {
133             return const SizedBox(height: 80);
```

7.1. EVIDENCIA:

```
93
94     if (snapshot.connectionState == ConnectionState.waiting && !snapshot.hasData) {
95         | return const Center(child: CircularProgressIndicator());
96     }
97
98     // Ordenar client-side por fechaInicio descendente
99     final chats = (snapshot.data?.docs ?? []).sort((a, b) {
100         | final dataA = a.data() as Map<String, dynamic>;
101         | final dataB = b.data() as Map<String, dynamic>;
102         | final tsA = dataA['fechaInicio'] as Timestamp?;
103         | final tsB = dataB['fechaInicio'] as Timestamp?;
104         | if (tsA == null && tsB == null) return 0;
105         | if (tsA == null) return 1;
106         | if (tsB == null) return -1;
107         | return tsB.compareTo(tsA);
108     });
109
110     if (chats.isEmpty) {
111         | return _buildEstadoVacio();
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113
114     return ListView.builder(
115         padding: const EdgeInsets.symmetric(vertical: 8, horizontal: 12),
116         itemCount: chats.length,
117         itemBuilder: (context, i) {
118             | final chat = chats[i];
119             | final data = chat.data() as Map<String, dynamic>;
120
121             | final isPublicador = data['publicadorId'] == currentUser.uid;
122             | final otherUserId =
123             | | isPublicador ? data['usuarioId'] : data['publicadorId'];
124             | final tipo = data['tipo'] ?? 'reporte';
125
126             | return FutureBuilder<DocumentSnapshot>(
127             | | future: FirebaseFirestore.instance
128             | | | .collection('usuarios')
129             | | | .doc(otherUserId)
130             | | | .get(),
131             | | builder: (context, userSnap) {
132             | | | if (!userSnap.hasData) {
133             | | | | return const SizedBox(height: 80);
```

```
134         }
135
136         final userData =
137         | | userSnap.data!.data() as Map<String, dynamic>? ?? {};
138         final nombre = userData['nombre'] ?? 'Usuario';
139         final fotoPerfil = userData['fotoPerfil'];
140
141         return StreamBuilder<QuerySnapshot>(
142             stream: FirebaseFirestore.instance
143             | .collection('chats')
144             | .doc(chat.id)
145             | .collection('mensajes')
146             | .orderBy('fechaEnvio', descending: true)
147             | .limit(1)
148             | .snapshots(),
149             builder: (context, msgSnap) {
150                 String ultimoMensaje = '';
151                 Timestamp? fechaUltimo;
152                 bool hayMensaje = false;
153
154                 if (msgSnap.hasData && msgSnap.data!.docs.isNotEmpty) {
155                     | final msg = msgSnap.data!.docs.first.data()
156                     | | as Map<String, dynamic>;
157                     ultimoMensaje = msg['texto'] ?? '';
158                     fechaUltimo = msg['fechaEnvio'] as Timestamp?;
159                     hayMensaje = true;
160                 }
161
162                 // Contar mensajes no leídos del otro usuario
163                 return StreamBuilder<QuerySnapshot>(
164                     stream: FirebaseFirestore.instance
165                     | .collection('chats')
166                     | .doc(chat.id)
167                     | .collection('mensajes')
168                     | .where('emisorId', isEqualTo: otherUserId)
169                     | .where('leido', isEqualTo: false)
170                     | .snapshots(),
171                     builder: (context, unreadSnap) {
172                         final noLeidos =
173                         | | unreadSnap.data?.docs.length ?? 0;
174
```

7.1. EVIDENCIA:

```
174
175         return _buildChatFile(
176             context: context,
177             chatId: chat.id,
178             data: data,
179             nombre: nombre,
180             fotoPerfil: fotoPerfil,
181             otherUserId: otherUserId,
182             tipo: tipo,
183             ultimoMensaje: ultimoMensaje,
184             fechaUltimo: fechaUltimo,
185             hayMensaje: hayMensaje,
186             noLeidos: noLeidos,
187             currentUid: currentUser.uid,
188         );
189     },
190 );
191 },
192 );
193 },
194 );
195 },
196 );
197 },
198 },
199 );
200 }
```

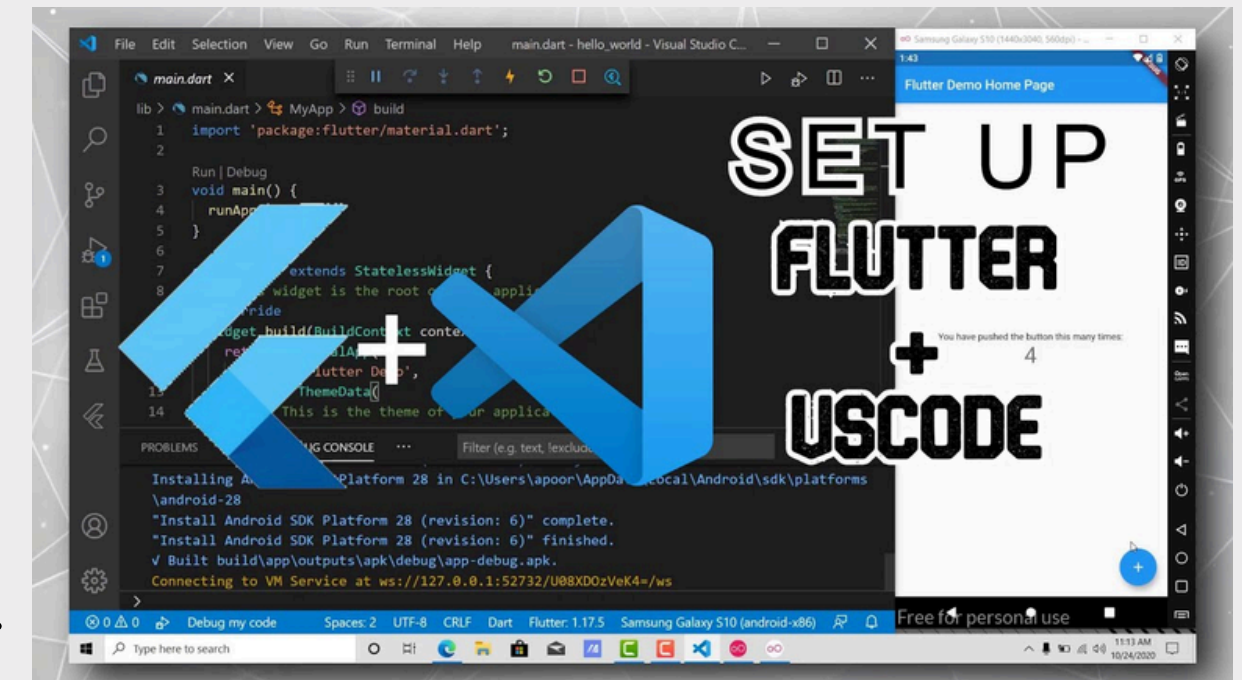
```
202 Widget _buildChatFile({
203     required BuildContext context,
204     required String chatId,
205     required Map<String, dynamic> data,
206     required String nombre,
207     required dynamic fotoPerfil,
208     required String otherUserId,
209     required String tipo,
210     required String ultimoMensaje,
211     required Timestamp? fechaUltimo,
212     required bool hayMensaje,
213     required int noLeidos,
214     required String currentUid,
215 }) {
216     ImageProvider? avatarImage;
217     if (fotoPerfil != null && fotoPerfil.isNotEmpty) {
218         avatarImage = NetworkImage(fotoPerfil);
219     } else {
220         avatarImage = AssetImage(obtenerAvatar(otherUserId));
221     }
222
223     return Container(
224         margin: const EdgeInsets.only(bottom: 10),
225         decoration: BoxDecoration(
226             color: Colors.white,
227             borderRadius: BorderRadius.circular(16),
228             boxShadow: [
229                 BoxShadow(
230                     color: Colors.black.withOpacity(0.05),
231                     blurRadius: 8,
232                     offset: const Offset(0, 3),
233                 ),
234             ],
235     );
```

8. RECURSOS

- Equipo humano
- Chite Quispe Brian Danilo (2021070015) - Desarrollo, análisis y pruebas.
- Anampa Pancca David Jordan (2022074268) - Desarrollo, análisis y pruebas.
- Ing. Ricardo Eduardo Valcarcel Alvarado - Docente asesor del curso Construcción de Software II.

RECURSOS TECNOLÓGICOS

- Flutter SDK 3.32, Android Studio 2024.2.1, Visual Studio Code.
- Firebase: Authentication, Firestore, Storage y Cloud Messaging (FCM v1).
- Modelos de IA embebidos con TensorFlow Lite (animales.tflite, extractor_animales.tflite) para detección y comparación de imágenes.
- Mapas e integración geográfica: flutter_map con OpenStreetMap Tile Server y geocodificación Nominatim.
- Entornos de prueba: Local, ALPHA y Beta, con dispositivos Android reales para pruebas piloto con usuarios.
- Control de versiones (Git) y herramientas de generación de pruebas asistida.



9. CRONOGRAMA

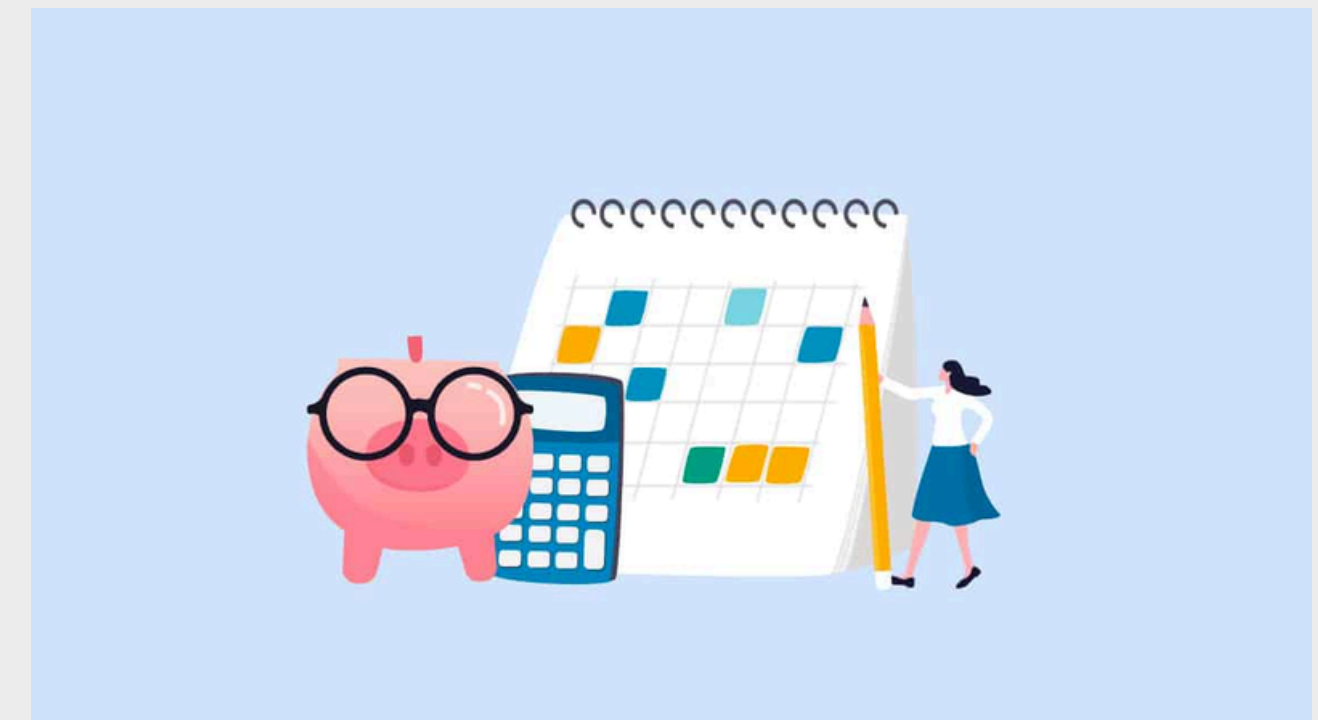
Fase 1 - Análisis y diseño (Construcción de Software I)

- 12 de abril de 2026 – 05 de julio de 2026: levantamiento de requerimientos (SRS) y diseño arquitectónico (SAD). Estado: completado.

Fase 2 - Construcción (Construcción de Software II)

- 30 de agosto de 2026 – 04 de diciembre de 2026: desarrollo de funcionalidades, revisión de código, catálogo y ejecución de pruebas, elaboración de trazabilidad, capacitación e implementación. Estado: en curso; este informe corresponde al corte del 18 de septiembre de 2026.
- Desarrollo de funcionalidades (RF-01 a RF-17).
- Revisión y refactorización de código (principios DRY, Fail Fast, nombres descriptivos, entre otros).
- Pruebas funcionales, unitarias, de integración, no funcionales, de regresión, de humo, de aceptación y punto a punto.
- Corrección de incidencias detectadas (incluye el hallazgo del módulo de chat).
- Implementación / despliegue.
- Capacitación a usuarios piloto.
- Puesta en producción y cierre del curso.

Fecha estimada de finalización: 04 de diciembre de 2026.



10. PRESUPUESTO

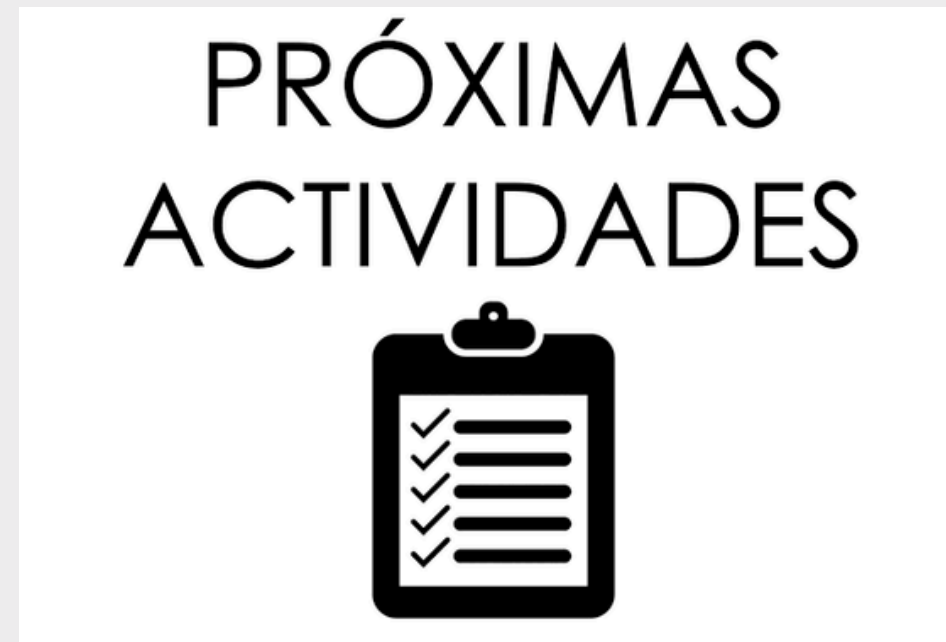
La siguiente tabla es una estimación de referencia elaborada a partir de los servicios y herramientas identificados en nuestro proyecto, debido a que son servicios gratuitos hasta que los despluguemos y activamos estos servicios con mejor recurso.

Concepto	Monto estimado	Observación
Servicios en la nube (Firebase/gratuito, uso actual de desarrollo y pruebas)	S/ 0.00	Sin costo mientras no se exceda la cuota gratuita.
Servicios en la nube proyectados en producción (Firebase plan Blaze: Firestore, Storage, FCM a mayor escala)	S/ 300.00 (estimado)	Estimación de referencia; depende del volumen real de usuarios.
Mapas y geolocalización (OpenStreetMap + Nominatim, uso actual)	S/ 0.00	Gratuito; evaluar migración a Google Maps API según el SAD (ver riesgo, sección 7).
Herramientas de desarrollo (Flutter SDK, Android Studio, VS Code, Git)	S/ 0.00	Software libre / licencias educativas.
Equipos (laptops propias del equipo)	S/ 0.00	Recursos ya disponibles del equipo.
Contingencia (10% sobre estimado de producción)	S/ 30.00	Reserva para imprevistos de servicios cloud.

PRESUPUESTO ESTIMADO TOTAL: S/ 330.00 (PROYECCIÓN PARA ESCENARIO DE PRODUCCIÓN).
PRESUPUESTO EJECUTADO: S/ 0.00 (USO EXCLUSIVO DE PLANES GRATUITOS).
PRESUPUESTO RESTANTE: NO APLICA AL NO HABERSE REGISTRADO EJECUCIÓN DE GASTO FORMAL.

11. PRÓXIMAS ACTIVIDADES

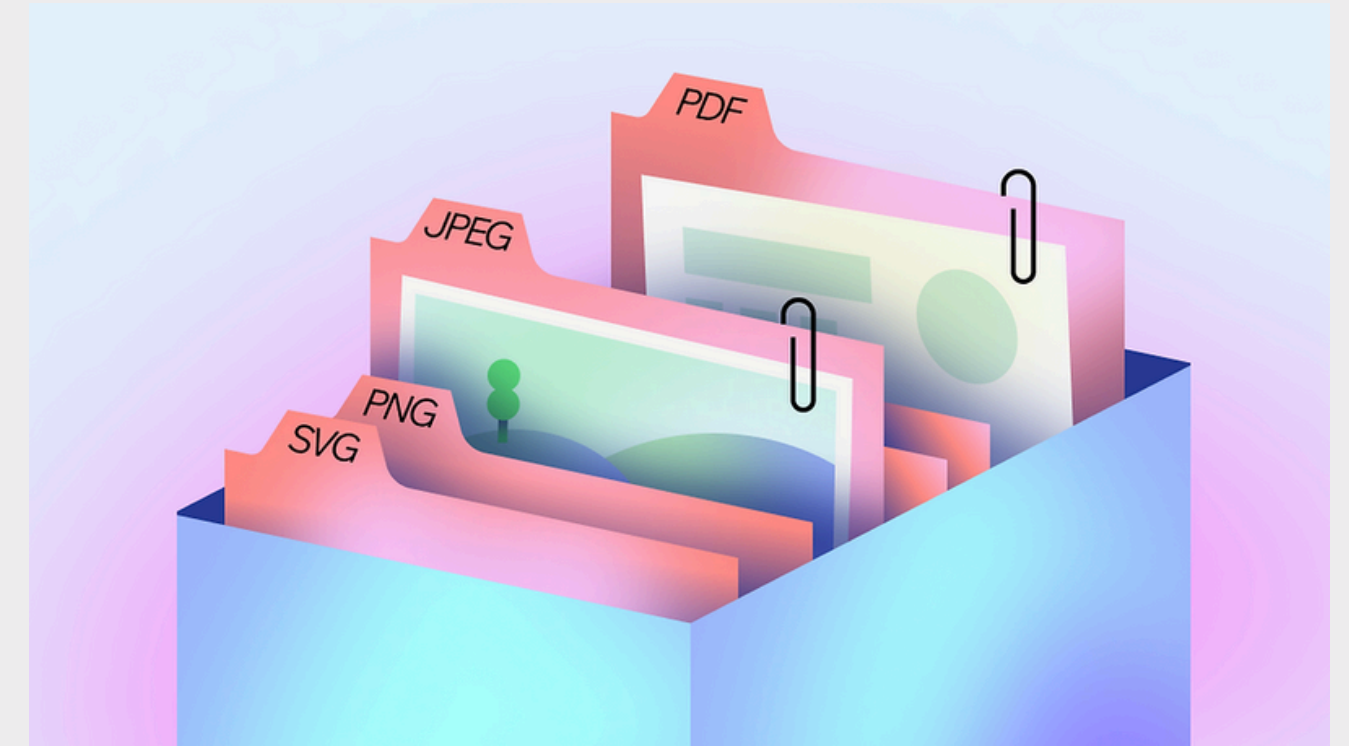
- Ejecutar las 87 pruebas pendientes del catálogo (integración, no funcionales, regresión, humo y punto a punto) conforme avance el desarrollo.
- Continuar la revisión y refactorización de código para los módulos aún no evaluados.
- Elaborar el manual de usuario y el material de capacitación.
- Preparar el ambiente de producción (evaluar paso a Firebase plan Blaze) y definir el plan de despliegue.
- Programar sesiones de capacitación y puesta en marcha antes del cierre del curso (04/12/2026).



12. CONCLUSIONES

El proyecto SOS Mascota presenta un avance general estimado de 45% y continúa orientado al cumplimiento de los objetivos definidos en el SRS y el SAD. Las fases de análisis y diseño se completaron en su totalidad durante Construcción de Software I, mientras que la fase de construcción (Construcción de Software II) avanza dentro del cronograma general del curso, aunque con una brecha relevante entre los 17 requerimientos funcionales especificados y los 5 que cuentan hoy con código productivo y pruebas aprobadas.

Se han identificado las acciones necesarias para gestionar los riesgos existentes, en particular el hallazgo sobre el módulo de chat (RF-08) y la necesidad de elevar la cobertura de pruebas antes del cierre. El siguiente hito relevante corresponde a completar el desarrollo funcional de los RF restantes y ejecutar las pruebas pendientes, lo que permitirá avanzar hacia la implementación, la capacitación y la puesta en producción programadas para el cierre del curso el 04 de diciembre de 2026.



ANEXOS

- Sección 5 (Avance del proyecto): captura de la corrida de pruebas automatizadas 93/93 (test/resultados_pruebas.txt) y del documento Pruebas_Consola_Evidencia.docx.
- Sección 7 (Riesgos): captura de pantalla_chat_activos.dart y de las rutas /chat y /chatsActivos mostrando el mensaje "Disponible en la próxima versión".
- Adjuntamos evidencia de los demás documentos de nuestro proyecto como el documento SAD (diagrama de componentes, despliegue, etc) y en la matriz de trazabilidad, para dar respaldo.
- Revision deCodigo: <https://docs.google.com/document/d/1QH-gio9e2MqodxT8f1W7ZGvAqslwtb-dSdmXmS9CtwY/edit?usp=sharing>
- Implementacion de Trazabilidad de implementación:
<https://docs.google.com/spreadsheets/d/15XOtl8GQ1cMe4oH0FhjsBV39ZmrhwKYr/edit?usp=sharing&ouid=116618994599667151200&rtpof=true&sd=true>
- Documento SAD: <https://docs.google.com/document/d/1KwHp7BskACG-K2bN8Q8MPYkTzSUUkzJq/edit?usp=sharing&ouid=116618994599667151200&rtpof=true&sd=true>
- Catálogo de Pruebas: <https://docs.google.com/spreadsheets/d/1D9k6gnWJ19HBf0mUH8pCpdDy5mpxYP/edit?usp=sharing&ouid=116618994599667151200&rtpof=true&sd=true>
- Documento de Factibilidad:
<https://docs.google.com/document/d/1mkMYz9lZJZolqrGEIvLXI3TcxLma2ovWRPJKIo12ZJg/edit?usp=sharing>

**MUCHAS
GRACIAS**